

Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 3

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

What's Inside?



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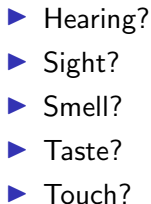
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What's Inside?



- ▶ Which of these seems the most valuable to you?
- ▶ Which would you relinquish last?

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Senses

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- ▶ Centralized
- ▶ Broad
- ▶ Passive
- ▶ Cognitive

Five Senses, Which is Most Important to You?

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Sight



- ▶ Centralized
- ▶ Broad
- ▶ Passive
- ▶ Cognitive



Touch



- ▶ Distributed
- ▶ Narrow
- ▶ Interactive
- ▶ Fundamental

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What is Haptics?

Definition

haptic

adjective

1. of or relating to the sense of touch, in particular relating to the perception and manipulation of objects using the senses of touch and proprioception.

origin

1. late 19th century: from Greek *haptikos* 'able to touch or grasp,' from *haptēin* 'fasten.'

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What is (Opt / Informat)ics?

Definition

optics

plural noun

1. the scientific study of sight and the behavior of light, or the properties of transmission and deflection of other forms of radiation.

Definition

informatics

plural noun

1. the science of processing data for storage and retrieval; informatics science

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Definition

Haptics

- ▶ the science and technology of touch

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Lateral Motion
Texture



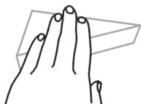
Pressure
Hardness



Enclosure
Global shape/Volume



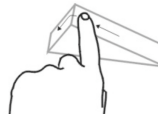
Static Contact
Temperature



Unsupported Holding
Weight



Contour Following
Shape



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Types of Sensing

Cutaneous / Tactile
Stems from skin contact



Kinesthetic / Proprioceptive
Stems from body movement

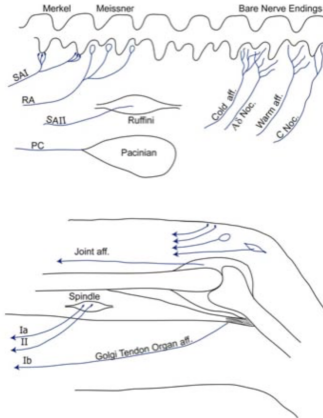


Illustration by A. M. Okamura.

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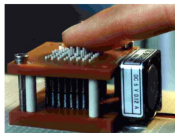
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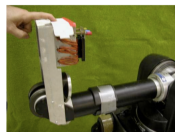
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Tactile Devices

Stimulate skin to create contact sensations

Haptic Interfaces



Hybrid Devices

Attempt to combine tactile and kinesthetic feedback



Kinesthetic Devices

Apply forces to guide or inhibit body movement

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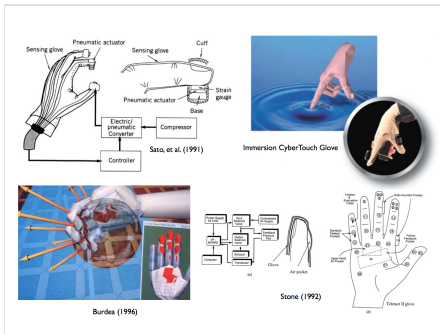
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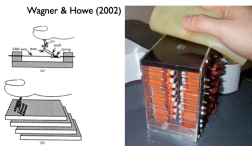
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Wagner & Howe (2002)



- ▶ Stimulation density needs to be high.
- ▶ Multimodal sensations are very hard to overlay.
- ▶ High complexity and weight diminish portability.
- ▶ Passive touch does not feel real.
- ▶ Tactile device design is difficult!

Kinesthetic Device Challenges

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Immersion CyberGrasp Glove



- ▶ Competing goals of high stiffness and low mass.
- ▶ Force feedback feels soft - “Nerfworld”.
- ▶ Point based interactions are overly simple.
- ▶ High quality devices are expensive.
- ▶ Limited workspace.
- ▶ Constrained to a desk!

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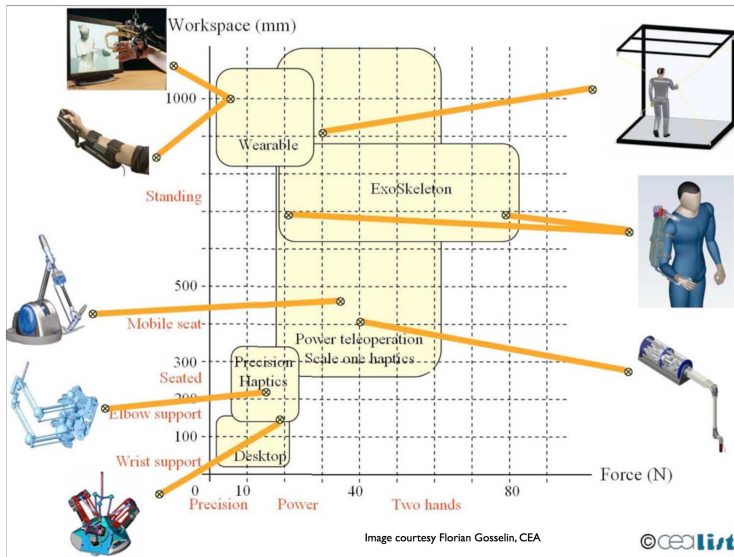
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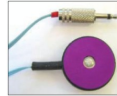
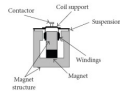
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But what can you do with haptic wearables?

- ▶ Vibration (eccentric mass motor or tactor)
- ▶ Skin stretch
- ▶ Kinesthetic force
- ▶ Inertial changes



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- ▶ OPTIONAL - FiDTouch - A 3D Wearable Haptic Display for the Finger Pad.pdf
- ▶ OPTIONAL - Haptics in Teleoperated Medical Interventions - Force Measurement, Haptic Interfaces and Their Influence on Users Performance.pdf
- ▶ OPTIONAL - Music Tactalizer - A Wearable Haptic Music Player with Multi-Feature Audio-Tactile Rendering.pdf
- ▶ OPTIONAL - Predicting the Future with Wearable Technology.pdf
- ▶ OPTIONAL - The Future of Wearable Technologies and Remote Monitoring in Health Care.pdf
- ▶ OPTIONAL - Ultralight Soft Wearable Haptic Interface with Shear-Normal-Vibration Feedback.pdf

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- ▶ **TO DISCUSS - Haptic Perception and Its Relation to Action.pdf**
- ▶ **TO DISCUSS - Haptic Rendering Introductory Concepts.pdf**
- ▶ **TO DISCUSS - The future of wearable technologies.pdf**